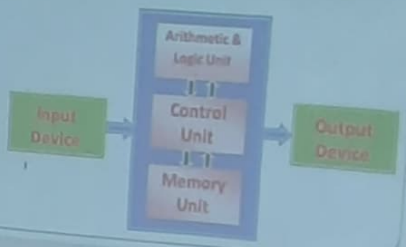
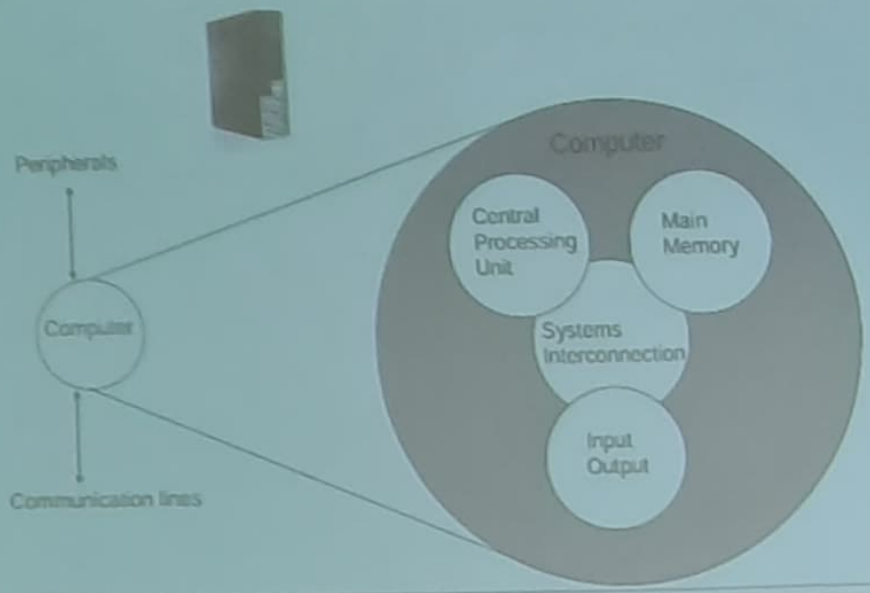
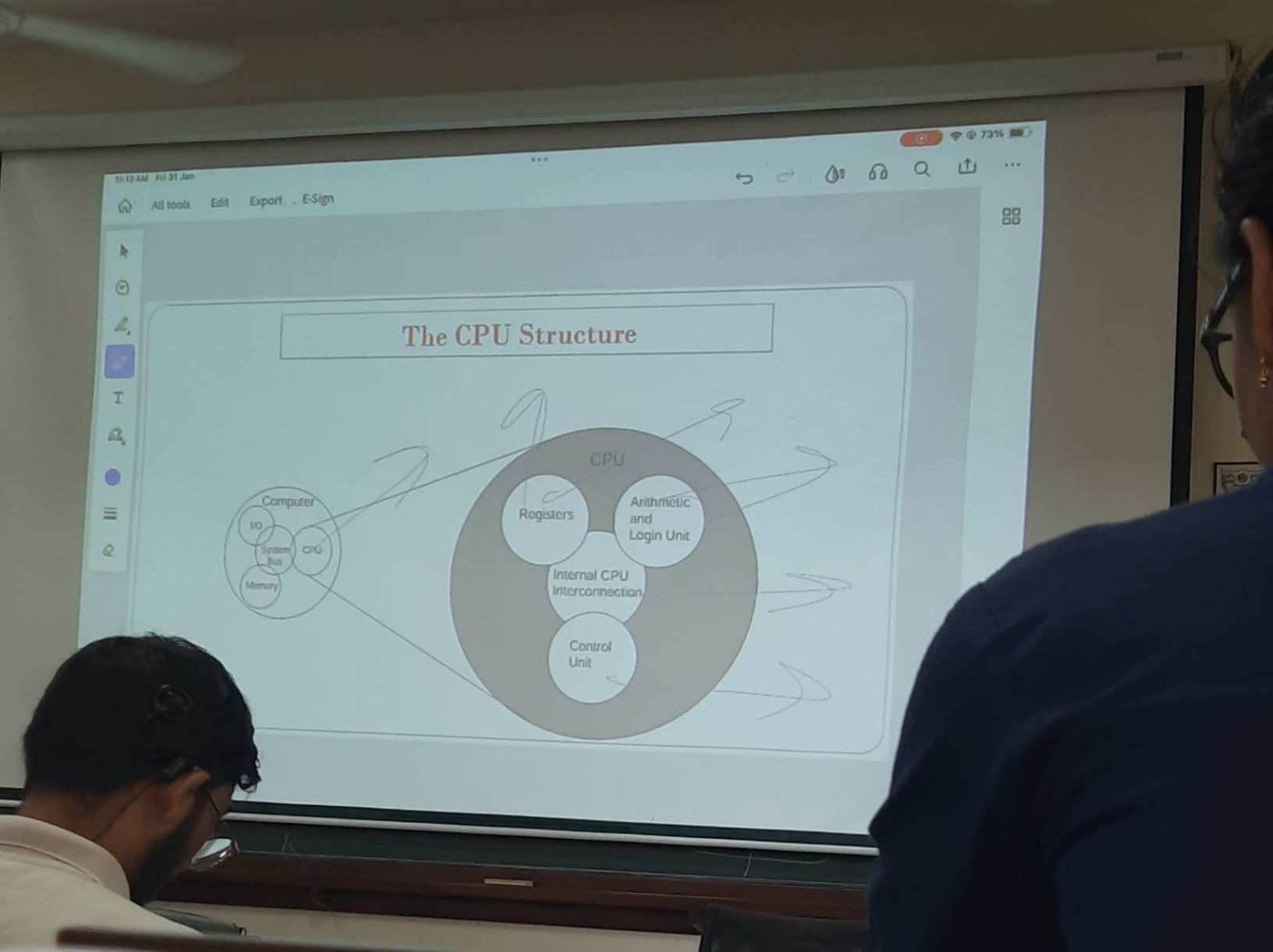
Computer System Components

- CPU/Microprocessor
- Memory Subsystem
- I/O Subsystem



Computer Structure - Top Level

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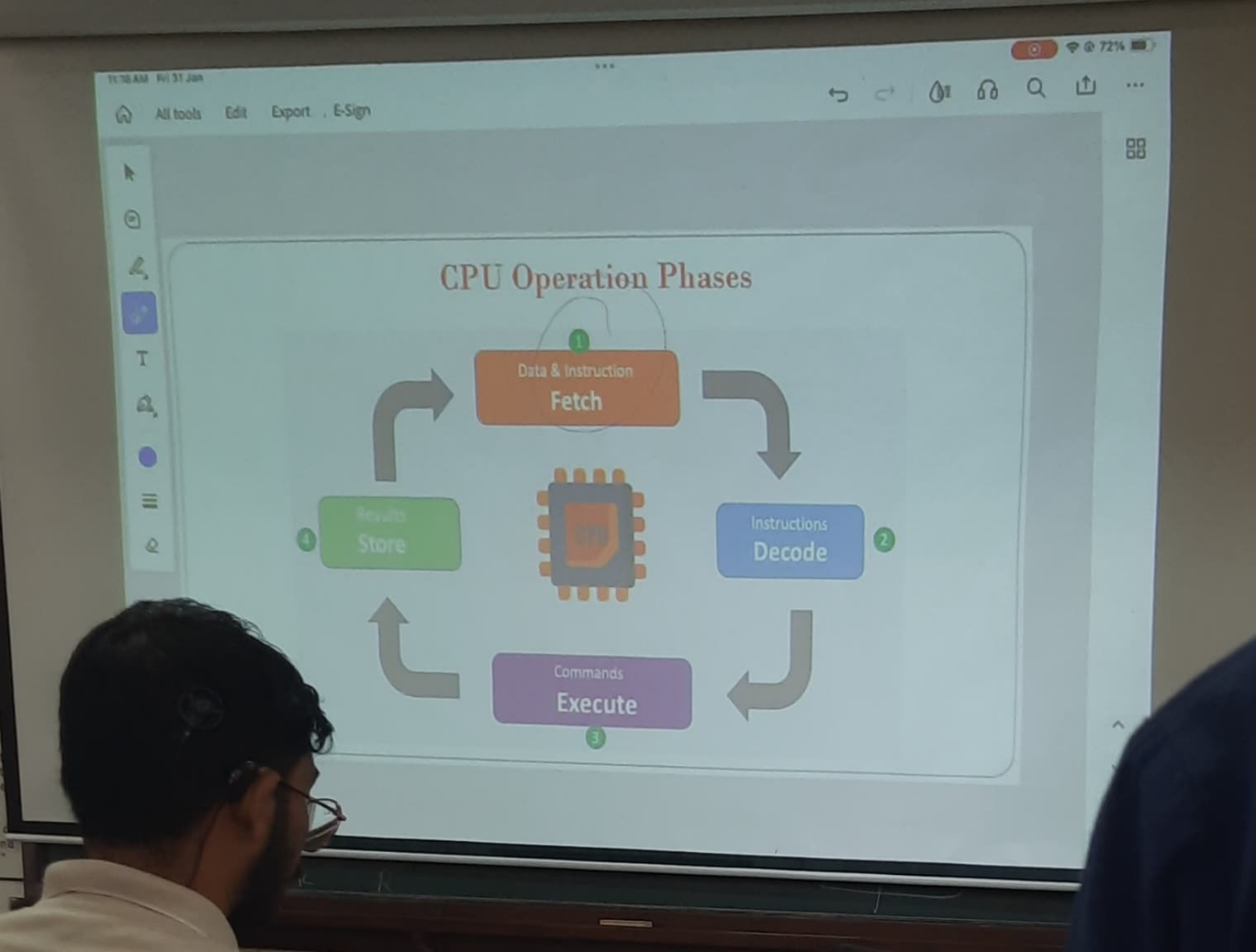
The CPU Structure
Computer
I/O
System Bus
CPU
Memory
CPU
Registers
Arithmetic and Login Unit
Internal CPU Interconnection
Control Unit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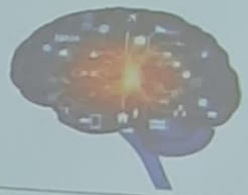
CPU Operation Phases
1
Data & Instruction
Fetch
2
Instructions
Decode
3
Commands
Execute
4
Results
Store
CPU

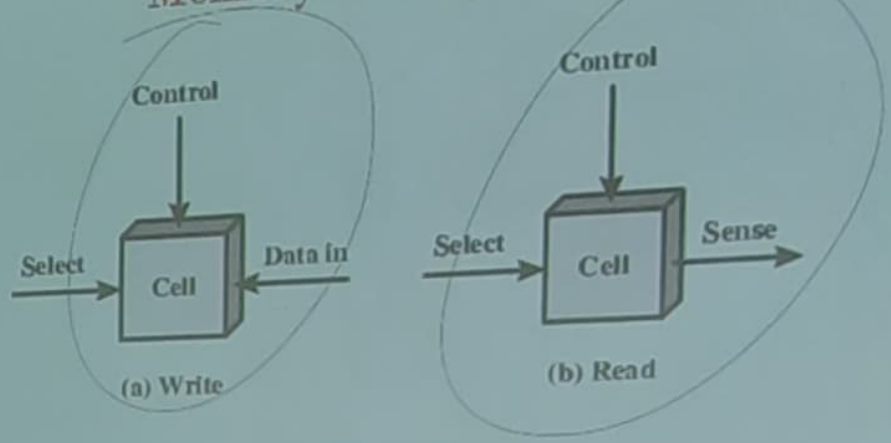
All tools Edit Export . E-Sign

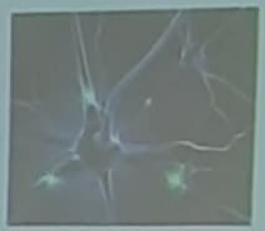
Memory

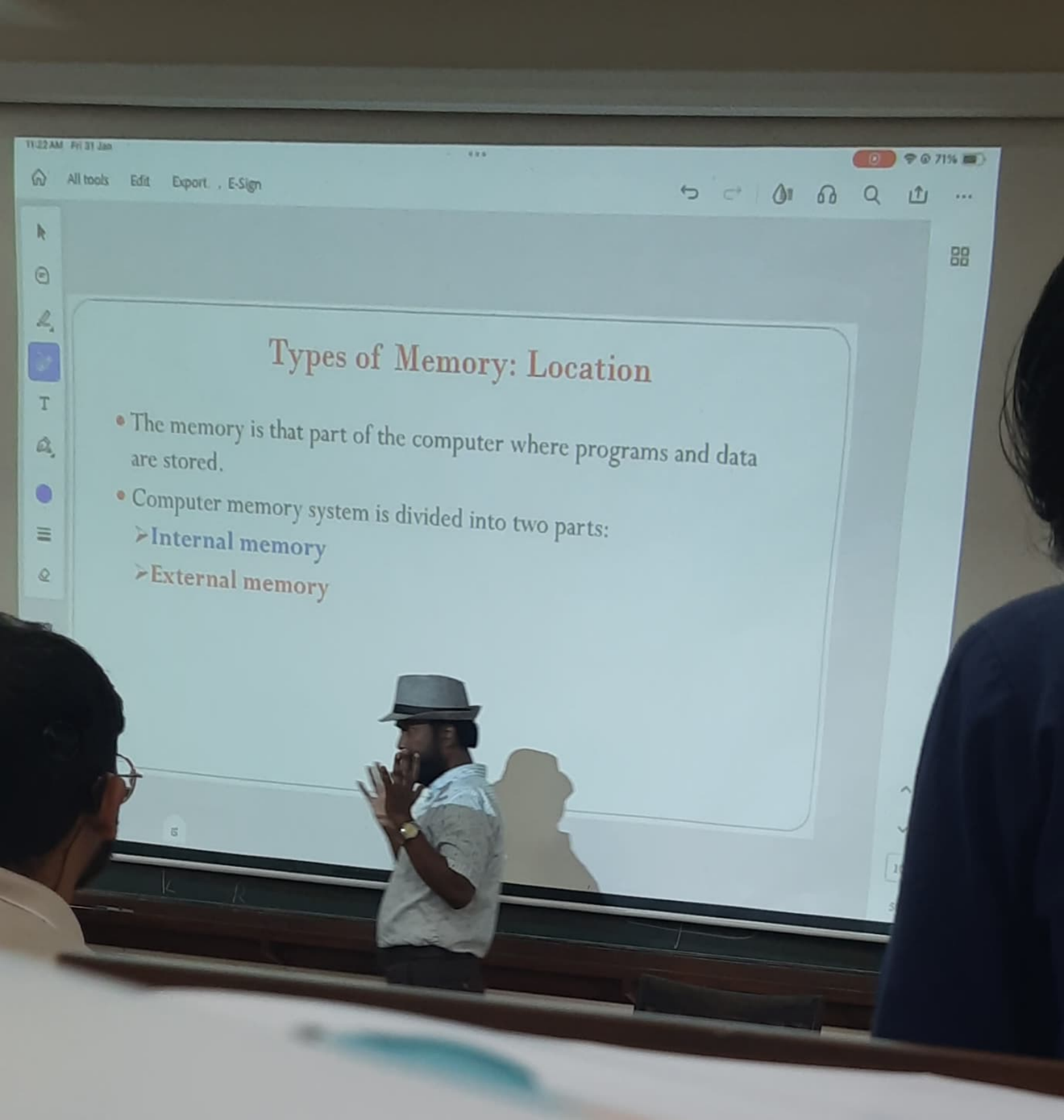
- *Computer memory is the storage space in the computer, where data is to be processed and instructions required for processing are stored.*
- The memory is divided into large number of small parts called cells.
- Each memory cell can hold one bit of information.
- Memory cells are organized in the form of an array.
- One row is one memory word.
- Passive component which stores data or instructions, accessed by address.
- Data flows from memory ("read") or to memory ("write").



Memory Cell Operation



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Types of Memory: Location
The memory is that part of the computer where programs and data are stored.
Computer memory system is divided into two parts:
Internal memory
External memory

Internal Memory Classification

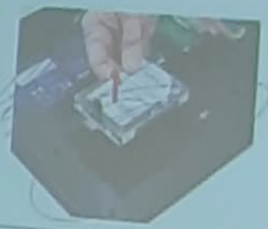
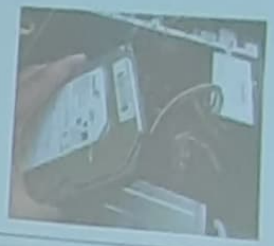
- According to Volatility:
  - RAM
  - ROM
- According to Technology:
  - Static RAM
  - Dynamic RAM

Internal Memory

- Data / instruction stored inside this storage can be accessible easily by CPU.
- It's a part of main component of a computer system
- Types : register, cache memory and main memory

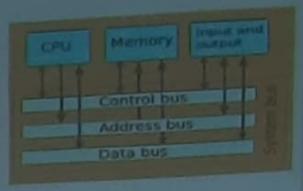
External Memory

- Data / instruction stored inside this storage cannot be accessible easily by CPU, must go through I/O Module or I/O unit
- It's a not an integral part of main component of a computer system
- Examples: magnetic disks, optical disks etc.

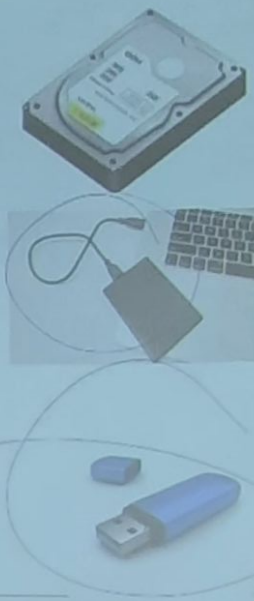
Primary Memory

- Primary storage (or main memory or internal memory), often referred to simply as memory, is the only one directly accessible to the CPU.
- The CPU continuously reads instructions stored there and executes them as required.
- Main memory is directly or indirectly connected to the CPU via a *memory bus*.
- It is actually two buses (not on the diagram): an address bus and a data bus.
- The CPU firstly sends a number through an address bus, a number called memory address, that indicates the desired location of data.
- Then it reads or writes the data itself using the data bus.
- is divided into RAM and ROM.



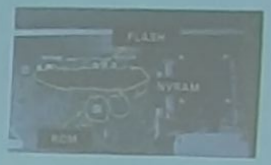
Secondary Memory

- The computer usually uses its input/output channels to access secondary storage and transfers the desired data using intermediate area in primary storage.
- Secondary storage does not lose the data when the device is powered down—it is non-volatile.
- Per unit, it is typically also an order of magnitude less expensive than primary storage.
- The secondary storage is often formatted according to a file system format, which provides the abstraction necessary to organize data into files and directories, providing also additional information (called metadata) describing the owner of a certain file, the access time, the access permissions, and other information.
- Hard disk are usually used as secondary storage.



Hybrid Memory

- As memory technology has matured in recent years, the line between RAM and ROM has blurred.
- Now, several types of memory combine features of both.
- These devices do not belong to either group and can be collectively referred to as hybrid memory devices.
- Hybrid memories can be read and written as desired, like RAM, but maintain their contents without electrical power, just like ROM.
- Two of the hybrid devices, EEPROM and flash, are descendants of ROM devices.
- These are typically used to store code. The third hybrid, NVRAM, is a modified version of SRAM.
- NVRAM usually holds persistent data.



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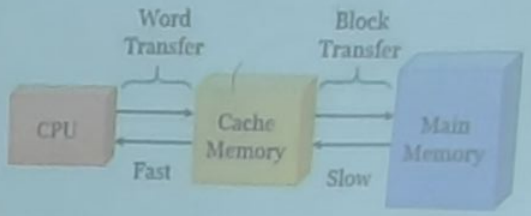
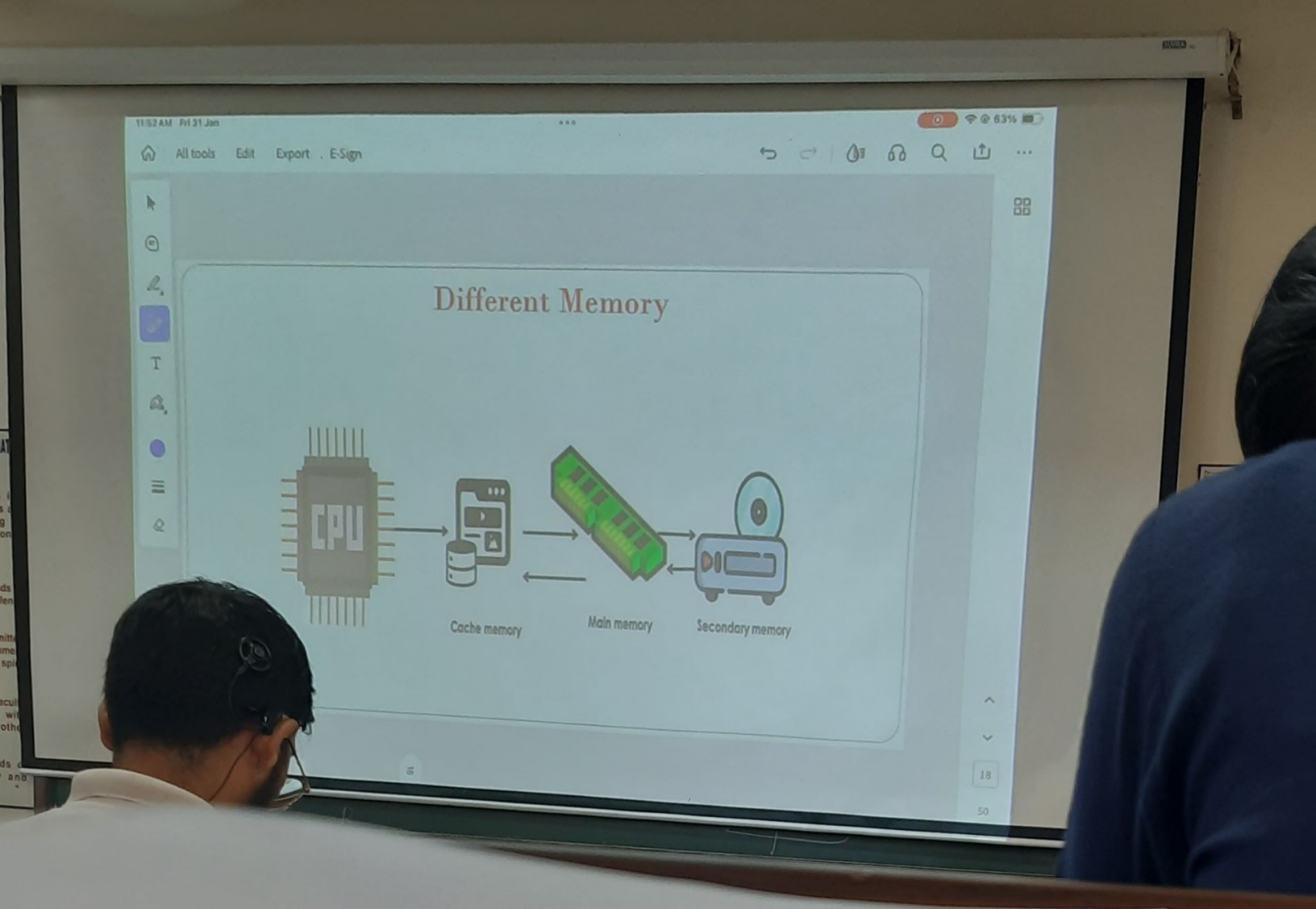
Cache Memory

- A CPU cache is a cache used by the central processing unit of a computer to reduce the average time to access memory.
- The cache is a smaller, faster memory which stores copies of the data from the most frequently used main memory locations.
- As long as most memory accesses are cached memory locations, the average latency of memory accesses will be closer to the cache latency than to the latency of main memory.
- When the processor needs to read from or write to a location in main memory, it first checks whether a copy of that data is in the cache.
- If so, the processor immediately reads from or writes to the cache, which is much faster than reading from or writing to main memory

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All tools
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Different Memory
CPU
Cache memory
Main memory
Secondary memory
18
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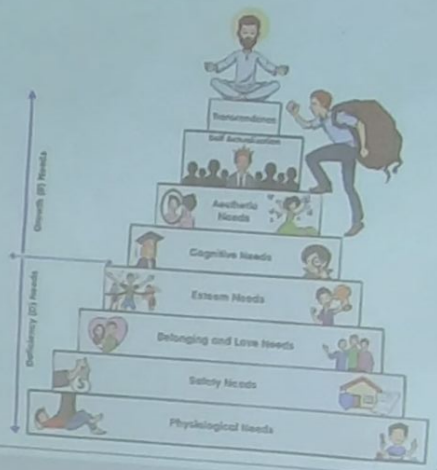
Memory Hierarchy

- *The Bottom Line:* there is a trade-off among the three key characteristics of memory.
  a) Capacity.
  b) Access time.
  c) Cost.



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Memory Hierarchy

- The way out of this dilemma is not to rely on a single memory component or technology, but to employ a memory hierarchy.
- The design constraints on a computer's memory can be summed up by three questions:

a) How much? → Capacity

- The question of how much is somewhat open ended.
- If the capacity is there, applications will likely be developed to use it.

b) How fast? → Time is money

- The question of how fast is, in a sense, easier to answer.
- To achieve greatest performance, the memory must be able to keep up with the processor.
- As the processor is executing instructions, we would not want it to have to pause waiting for instructions or operands.

c) How expensive? → Cost

- For a practical system, the cost of memory must be reasonable in relationship to other components.

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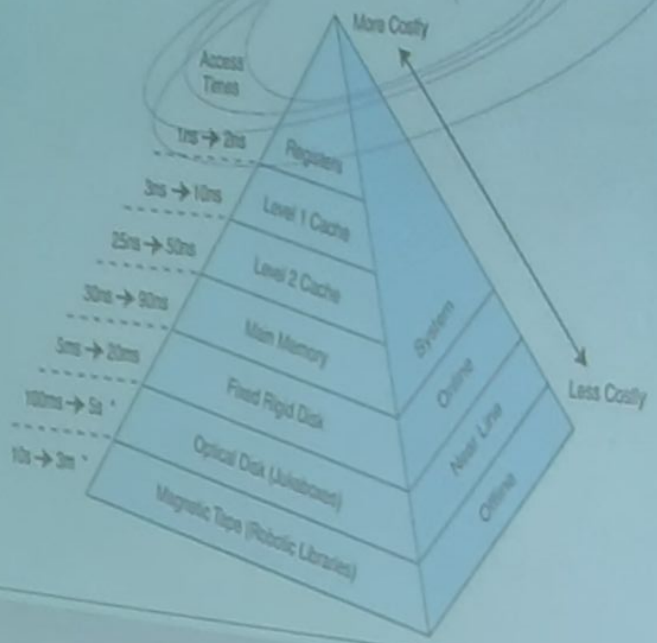
Memory Hierarchy

- A variety of technologies are used to implement memory systems, and across this spectrum of technologies, the following relationships hold:
  a) *Faster access time, greater cost per bit*
  b) *Greater capacity, smaller cost per bit*
  c) *Greater capacity, slower access time*
- The designer would like to use memory technologies that provide for large-capacity memory, both because the capacity is needed and because the cost per bit is low.
- However, to meet performance requirements, the designer needs to use expensive, relatively lower-capacity memories with short access times.

Memory Hierarchy- Pyramid



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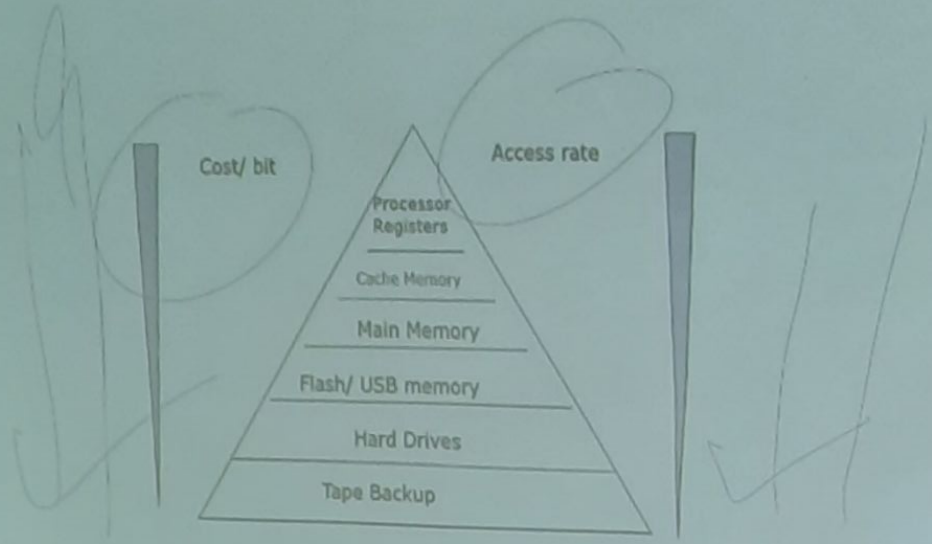
Memory Hierarchy

- As one goes down the hierarchy, the following occur:
  - a) Decreasing cost per bit
  - b) Increasing capacity
  - c) Increasing access time
  - d) Decreasing frequency of access of the memory by the processor

Memory Hierarchy



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3-Level Memory Classification

A. **Inbound storage / memory**
- Registers, Cache, Main memory.
- The class of Internal memory

B. **Outbound Storage**
- Magnetic Disk
- CD ROM
- CD RW
- DVD + RW
- DVD + RAM
- The class of External memory

C. **Off-line Storage**
- Magnetic Tape
- [illegible]d Disk Drive
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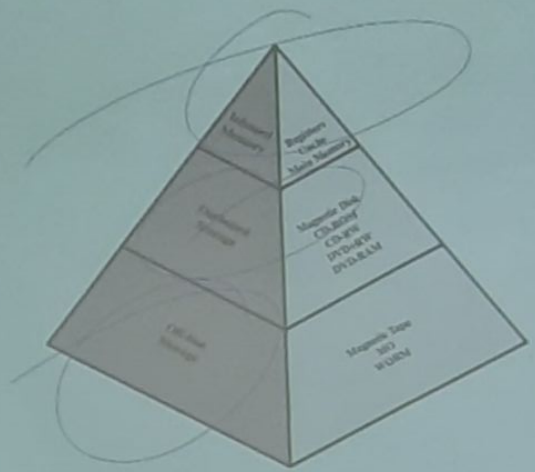
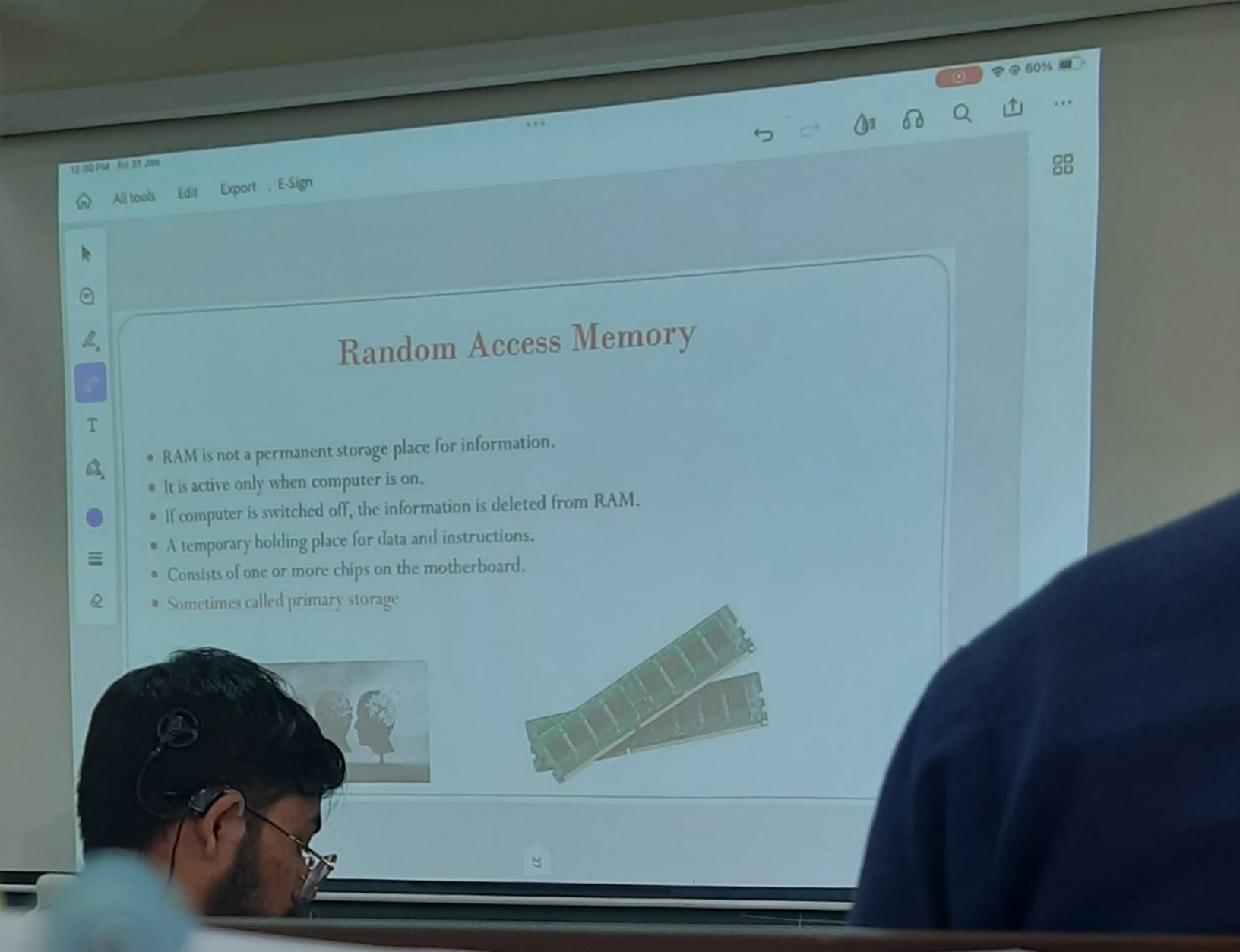
Memory Hierarchy - Diagram

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Random Access Memory
RAM is not a permanent storage place for information.
It is active only when computer is on.
If computer is switched off, the information is deleted from RAM.
A temporary holding place for data and instructions.
Consists of one or more chips on the motherboard.
Sometimes called primary storage
27

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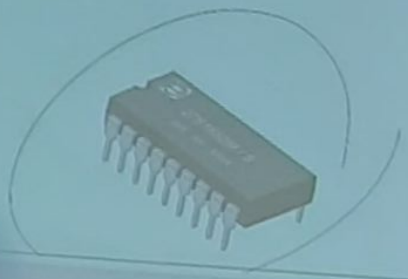
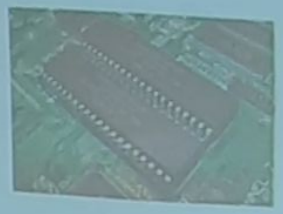
Read Only Memory

- ROM is the memory where information is only read.
- When the computer is turned a off, information is not deleted from ROM.

- Writing information to ROM is made by vendor.
  - Mostly used for installation into toys or appliances
  - Programmed once although there are exceptions
    - PROM/EPROM





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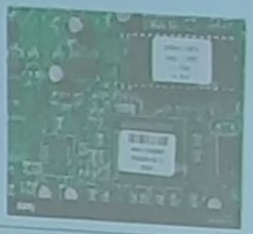
Read Only Memory

- Permanent storage
  - Nonvolatile
- Microprogramming
- Library subroutines
- Systems programs (BIOS)
- Function tables



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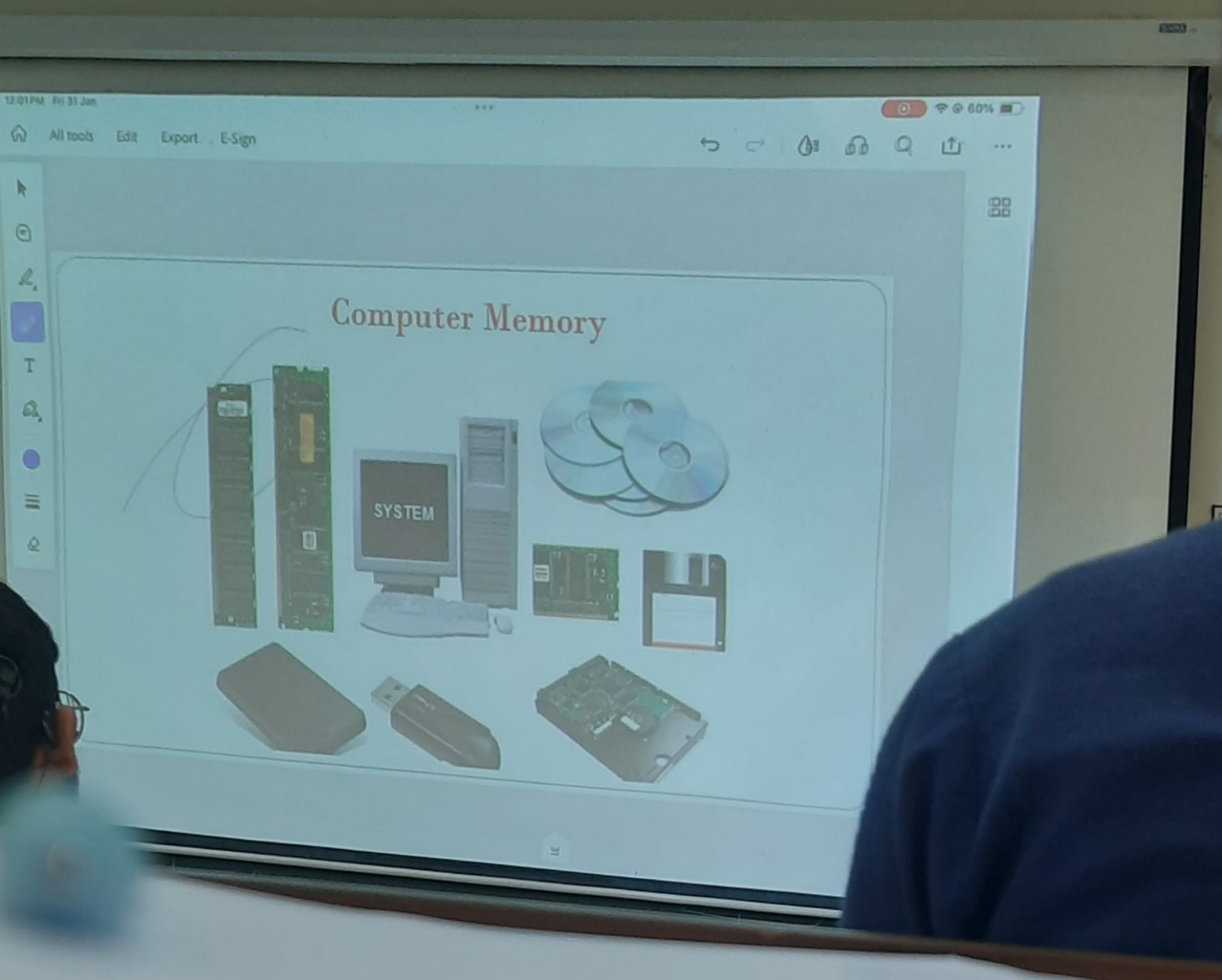
Types of ROM

- Written during manufacture
  - Very expensive for small runs
- Programmable (once)
  - PROM
  - Needs special equipment to program
- Read "mostly"
  - Erasable Programmable (EPROM)
    - Erased by UV
  - Electrically Erasable (EEPROM)
    - Takes much longer to write than read
  - Flash memory
    - Erase whole memory electrically, no byte-level erase
    - But much faster write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Computer Memory
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